

LINDEN AGCM, AB, Canada

WMO: 710350

Lat: 51.6192N

Lon: 113.6559W

Elev: 2999

StdP: 13.17

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.9	-19.6	-33.0	1.0	-26.9	-25.3	1.5	-19.4	26.9	25.0	22.7	31.5	3.5	100	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	83.2	61.2	79.9	60.1	76.8	59.3	65.8	77.3	63.7	75.0	61.8	72.7	8.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	90.8	72.0	59.2	84.0	69.7	57.1	77.8	67.5	32.3	77.6	30.6	75.1	29.1	72.6	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.9	20.3	17.7	DB	-36.3	90.3	5.4	3.8	-40.2	93.1	-43.4	95.3	-46.4	97.5	-50.4	100.3	(4)
(5)			WB	-36.5	70.5	5.4	2.6	-40.4	72.3	-43.5	73.8	-46.6	75.3	-50.5	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	14.7	12.9	24.3	36.9	49.1	56.1	60.1	58.9	50.8	38.1	23.7	11.6		
	DBStd	20.79	15.04	15.12	14.50	8.98	7.40	5.12	4.34	5.15	8.03	8.85	13.70	15.04		
	HDD50	5890	1094	1039	796	398	108	10	0	3	84	378	790	1190		
	HDD65	10398	1559	1459	1261	844	493	269	161	197	428	833	1239	1655		
	CDD50	989	0	0	0	5	81	193	312	279	108	10	1	0		
	CDD65	20	0	0	0	0	0	2	8	8	2	0	0	0		
	CDH74	1325	0	0	0	9	116	131	372	445	242	10	0	0		
	CDH80	274	0	0	0	0	14	16	73	111	60	0	0	0		
	WSAvg	7.8	7.0	8.1	7.1	8.1	8.8	9.2	7.3	6.6	7.2	9.2	7.5	7.3		
	PrecAvg	20.3	0.7	0.6	1.0	1.5	2.7	3.9	2.9	2.5	1.8	1.0	0.9	0.8		
(15) Precipitation	PrecMax	26.7	1.5	1.3	2.1	3.4	4.3	10.0	6.3	4.5	4.3	2.0	2.4	1.7		
	PrecMin	12.9	0.2	0.1	0.3	0.6	0.9	1.7	1.1	0.5	0.3	0.3	0.2	0.2		
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.0	1.5	1.2	0.9	0.9	0.4	0.5	0.4		
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.0	1.5	1.2	0.9	0.9	0.4	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.9	47.6	59.9	74.0	82.0	82.4	86.0	88.4	86.4	74.3	62.9	47.7		
		MCWB	38.8	38.6	44.8	48.8	54.3	61.0	64.6	62.0	58.0	51.6	46.4	37.4		
	2%	DB	43.1	41.8	53.5	66.3	77.4	77.8	81.9	82.8	80.5	66.2	53.7	41.2		
		MCWB	36.4	35.4	40.7	47.2	52.9	58.6	64.6	61.9	56.6	48.6	42.2	35.1		
	5%	DB	40.1	37.9	48.1	59.9	72.5	74.1	78.1	79.1	76.0	61.2	47.0	38.1		
		MCWB	34.9	32.8	38.3	43.5	51.4	57.0	63.8	61.6	55.5	46.5	38.3	33.0		
	10%	DB	36.2	34.0	42.6	54.4	68.0	70.9	74.8	75.0	69.9	55.7	42.2	33.2		
		MCWB	32.0	30.5	35.8	40.8	50.1	55.5	61.8	60.5	53.0	44.1	35.4	29.5		
	0.4%	WB	39.3	39.3	45.5	50.8	55.9	65.0	70.0	67.1	60.6	53.9	47.1	38.5		
		MCDB	47.3	46.4	57.9	71.4	75.8	77.3	79.3	79.1	80.1	70.6	60.4	46.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	35.8	42.0	47.4	54.0	61.5	66.9	65.0	58.4	50.2	42.7	35.6		
		MCDB	42.8	41.3	51.7	64.8	73.1	73.4	78.1	77.1	76.1	63.1	52.5	41.0		
	5%	WB	34.9	33.2	38.6	44.4	52.5	59.0	64.6	63.2	56.2	47.4	38.9	32.8		
		MCDB	39.9	37.4	47.1	57.3	69.6	69.7	75.4	75.1	72.4	60.0	45.8	37.6		
	10%	WB	32.1	30.6	36.0	41.6	50.8	56.9	62.8	61.0	54.1	44.8	35.9	29.3		
		MCDB	36.2	34.0	42.3	52.5	65.5	67.8	73.3	72.7	69.6	54.8	41.4	33.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	22.3	21.7	24.6	26.9	24.1	26.3	28.5	29.2	27.0	22.8	21.1		
		MCDBR	22.1	25.4	29.1	37.9	37.7	32.3	34.1	38.1	42.0	36.2	30.0	24.8		
		MCWBR	16.4	19.3	17.6	19.8	16.3	15.3	19.0	19.2	19.4	20.0	19.0	18.7		
	5% WB	MCDBR	21.3	23.7	27.3	35.7	32.6	27.5	31.0	32.5	37.6	34.7	28.6	24.5		
		MCWBR	16.3	18.3	17.1	18.9	14.3	15.0	18.5	18.2	17.9	19.3	19.0	18.9		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.272	0.283	0.331	0.342	0.344	0.344	0.359	0.313	0.281	0.253	0.239		
	taud		2.359	2.407	2.459	2.347	2.374	2.405	2.409	2.370	2.458	2.531	2.479	2.362		
	Ebn at Noon		246	276	294	288	289	289	286	276	279	270	251	235		
	Edh at Noon		19	24	28	35	36	35	35	34	28	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		348	670	1104	1494	1801	1922	2057	1673	1177	712	379	272		
	RadStd		25	54	89	119	122	121	107	118	126	82	30	25		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page